HDOC-65

Recursive Program - Tower of Hanoi

Question

Write a recursive program to solve the Tower of Hanoi problem for n disks and display all moves.

Aim

To implement the Tower of Hanoi problem using recursion and display the sequence of moves required to transfer all disks from the source rod to the destination rod.

Algorithm / Logic

1. Read the number of disks n from the user.
2. Call the recursive function towerOfHanoi(n, source, auxiliary, destination).
3. If n = 1, move the disk directly from the source rod to the destination rod.
4. Otherwise, move n - 1 disks from the source rod to the auxiliary rod using the destination rod.
5. Move the nth (largest) disk from the source rod to the destination rod.
6. Move the n - 1 disks from the auxiliary rod to the destination rod using the source rod.
7. Continue recursively until all disks are moved.
8. The minimum number of moves required is 2^n - 1.

Function Used

towerOfHanoi(int n, char source, char auxiliary, char destination) - a recursive user-defined function that displays every valid disk movement.

Header File Used

#include <stdio.h> - used for standard input and output functions such as printf() and scanf().

C Program

```
#include <stdio.h>

void towerOfHanoi(int n, char source, char auxiliary, char destination)
{
   if (n == 1)
   {
      printf("Move disk 1 from %c to %c\n", source, destination);
      return;
   }

   towerOfHanoi(n - 1, source, destination, auxiliary);
   printf("Move disk %d from %c to %c\n", n, source, destination);
```

```
    towerOfHanoi(n - 1, auxiliary, source, destination);
}

int main()
{
    int n;

    printf("Enter the number of disks: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Number of disks must be positive.\n");
        return 0;
    }

    printf("\nSequence of moves:\n");
    towerOfHanoi(n, 'A', 'B', 'C');

    return 0;
}
```

Explanation

The Tower of Hanoi consists of three rods and n disks of different sizes. Initially, all disks are placed on the source rod with the largest disk at the bottom and the smallest at the top.

The program uses recursion. To move n disks from rod A to rod C, it first moves n - 1 disks from A to B, moves the largest disk from A to C, and then moves the n - 1 disks from B to C. The base case occurs when only one disk remains.

For n disks, the minimum number of moves is 2^n - 1. Therefore, for 3 disks, 7 moves are required.

Test Cases

Test Case	Input (n)	Expected Moves	Observation
1	1	1	Single disk moves directly from A to C.
2	2	3	All disks move correctly in the minimum 3 moves.
3	3	7	All disks move correctly in the minimum 7 moves.
4	4	15	Recursive solution generates the minimum 15 moves.

Sample Output

```
Enter the number of disks: 3

Sequence of moves:
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
```

```
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
```

Result

Thus, the Tower of Hanoi problem was successfully solved using recursion. The program displays all the moves required to transfer n disks from the source rod to the destination rod while following the rules of the Tower of Hanoi.